6 An electron is confined within a one-dimensional box of width L.

ψ represents the sinusoidal wavefunction of the electron. The graph of $|\psi|^2$ against distance x from one wall of the box is shown in Figure 6.1.

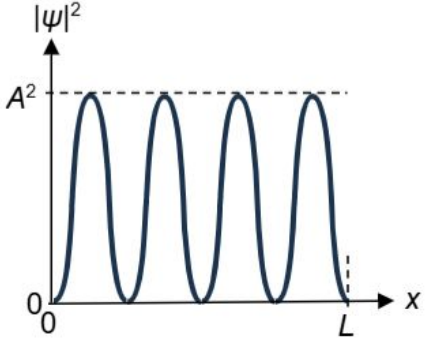


Figure 6.1

(a) (i) State what is represented by $|\psi|^2$.

.....

..... [1]

L1 **Probability density (OR Probability per unit length) of finding the position of the electron.** **B1**

(ii) The graph of $|\psi|^2$ against x in Figure 6.1 has an amplitude of A^2.

Derive an expression for A in terms of L.

			A = ...	[2]
		L1	Total Area under the graph = Total Probability = 1 $\int_0^L \|\psi\|^2 dx = 1$ $\frac{A^2}{2} L = 1$ $A = \sqrt{\frac{2}{L}}$	M1 A1
	(b)	The electron with the $\|\psi\|^2$ graph shown in Figure 6.1 is in an excited quantum state *n* in a box where *L* = 0.10 nm.		
		(i)	State the quantum state *n* of the trapped electron shown in Figure 6.1.	
			n = ...	[1]
		L2	n = 4	B1
		(ii)	Calculate the momentum of the electron at this quantum state.	
			momentum =..... N s	[2]
		L2	Wavelength, λ = L / 2 = 0.10 / 2 = 0.050 nm = 0.050 x 10^{-9} m $p = \frac{h}{\lambda} = \frac{6.63\times10^{-34}}{0.050\times10^{-9}}$ = 1.326 x 10^{-23} = **1.3 x 10^{-23} N s**	M1 A1
		(iii)	Calculate the energy, in eV, of the electron at this quantum state.	

			energy = ... eV	[2]
		L1	$E_n = \frac{n^2h^2}{8mL^2}$ $= \frac{(4)^2(6.63\times10^{-34})^2}{8(9.11\times10^{-31})(0.10\times10^{-9})^2}$ = 9.65025 x 10^{-17}J **= 603 eV** ***Allow ECF of value of n in part (b)(i).*** **OR use $E_n = \frac{p^2}{2m}$ where *p* is found in (b)(ii).**	**M1** **A1**

[Total: 8]